REPLICATION PACKAGE

Creative Destruction through Innovation Bursts

This replication package accompanies Berlingieri, De Ridder, Lashkari, and Rigo (forthcoming), "Creative Destruction through Innovation Bursts," *Econometrica*.

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NAVIGATION

Contents

Instructions to replicators

- Theoretical and quantitative analysis
- Empirical analysis

Computational requirements

- Software requirements
- Memory and runtime requirements

Data requirements

- Details on each data source
- Dataset list

Description of programs/code

- Description of theoretical programs
  - Parameters and quantification
  - Simulation and dataset construction
  - Theoretical tables and figures
- Description of empirical programs
  - SAS export programs
  - Dataset-construction programs
  - Empirical tables and figures

List of tables and figures

References

- Data references
- Other references

Instructions to replicators

The replication package is split into a `theory` folder, which reproduces the theoretical and quantitative results (section 4 and 5), and an `empirics` folder, which reproduces the empirical results (section 2 and 3).

The theoretical results can be reproduced on computers with the specifications described below. The empirical analysis can only be performed by researchers with access to the confidential French microdata in the CASD secure environment as described below.

Theoretical and quantitative analysis

The full theoretical replication can be performed by running `file1_master.m` from the `theory` folder.

Before running the file:

1. Open `file1_master.m` and change `stata_exe` to the location of the Stata executable on the local computer.
2. For a complete run from the model primitives, set the execution flags at the top of `file1_master.m` to:

```
quantification      = 1;
simulation          = 1;
calibrate_noise     = 1;
create_datasets     = 1;
output_tables       = 1;
run_stata_figures   = 1;
```

3. Make the `theory` folder the current MATLAB directory.
4. Run `file1_master.m`.

The master performs the following steps:

1. Loads the exogenous parameters and calibration targets.
2. Quantifies the model and saves the preferred calibration in `estimation_results.mat`.
3. Simulates 100 samples of 22,000 firms over 10 years for the innovation-bursts model and the Klette-Kortum benchmark.
4. Calibrates the standard deviation of idiosyncratic quality shocks to the Shapley-Owen target and saves the result in `estimated_noise_sd.mat`.
5. Converts the raw MATLAB simulations into non-proprietary firm- and product-level CSV datasets.
6. Produces Tables V and VI in MATLAB.
7. Generates the firm-size CSV inputs used by Stata.
8. Calls `stata_file1_master.do`, which produces Figures 9-12, Appendix Figure C13, the model results for Tables VII and VIII, and the simulated firm-age moments reported in the text.
9. Produces Figures 13 and 14 in MATLAB.

All tables, figures, and intermediate MATLAB-to-Stata files are saved in the `theory/plots` folder. The firm- and product-level simulation datasets are saved in:

```
theory/montecarlo/simulationoutput
theory/montecarlo/simulationoutput_kk
```

The raw MATLAB simulation folders are temporary. After each raw simulation has been converted to CSV, the master deletes the corresponding raw folder.

The replicator should expect the full theoretical routine to take approximately six hours on hardware comparable to that described below.

Empirical analysis

The empirical analysis can only be performed by researchers with access to the confidential French microdata in the CASD environment. Details on each data source and on how to obtain access are provided in the data-availability section below.

The full empirical replication can be performed by running `empirics/code/master.do`.

Before running the file:

1. Import the entire `empirics` folder into the CASD environment using the standard import procedure (imports-exports@casd.eu).
2. Open `empirics/code/master.do` and change the three paths defined at lines 35-37:
   - `global path`: the location of the `empirics` folder within the CASD environment.
   - `global sas`: the location of the SAS executable (`sas.exe`) on the CASD server.
   - `global raw`: the location of the raw CASD data folder, which has the format `//casd.fr/casdfs/Projets/PROJECTNAME/Data`, where `PROJECTNAME` is the name of the replicator's CASD project.

No other paths need to be changed: all other programs read and write relative to `global path`.

Then run `empirics/code/master.do` with Stata. The master performs the following steps:

1. Installs the user-written packages, and sets the seed and the graph scheme.
2. Creates the folder structure `empirics/data/raw`, `empirics/data/processed`, and `empirics/output`.
3. Calls the six SAS scripts in `empirics/code/sas` through Windows `shell` commands. These convert the raw SAS microdata into Stata datasets saved in `empirics/data/raw`.
4. Constructs the analysis datasets by running the programs in `empirics/code/data`. All intermediate datasets are saved in `empirics/data/processed`.
5. Runs the programs in `empirics/code/analysis`, which produce all empirical tables and figures, including the data columns of Tables VII and VIII and the empirical in-text numbers.
6. Runs `rename_output_files.do`, which renames the panels of the multi-panel figures to the final file names listed in the list of tables and figures below.

All empirical tables and figures are saved in `empirics/output`.

The global `conf_check` in `empirics/code/master.do` (set to 1 by default) implements the confidentiality treatment of the distribution plots described in Appendix B.5 of the paper. It must be set to 1 for any output that is exported from the CASD environment. The sample period is set by the globals `yr_start` (2010) and `yr_end` (2019).

Computational requirements

Software requirements

The **empirical code** was last run in the CASD environment on:

- Stata/MP 17.0.
- SAS 9.4, which is used only to convert the raw SAS microdata to Stata format. SAS is available within the CASD environment; the SAS scripts are called from Stata through Windows `shell` commands.

The following user-written Stata packages are required by the empirical code:

- `carryforward`, `shapley2`, `gtools`, `ftools`, `ppmlhdfe`, `reghdfe`, `winsor`, `boottest`, `outreg2`, `estout`, `binscatter`, and `fs`, which are installed automatically at the top of `empirics/code/master.do`.

The **theoretical code** was last run on:

- MATLAB R2024b, Update 3.
- Stata/MP 18.

The following MATLAB toolboxes are required:

- Optimization Toolbox.
- Statistics and Machine Learning Toolbox.
- Parallel Computing Toolbox.
- Symbolic Math Toolbox.

The theoretical code includes `randraw.m`. Its original three-clause BSD license notice is retained in the source file.

Memory and runtime requirements

Empirical analysis: The empirical code was last run on an 24-core server with 32 GB of RAM. Storage requirement: approximately 30 GB of free storage. Computation took approximately just under 3 hours.

Theory: The theoretical code was last run on an 8-core AMD Ryzen-based Windows desktop with 32 GB of RAM and processors running at approximately 4.7 GHz. The full routine takes approximately six hours. The generated Monte Carlo CSV files occupy approximately 11 GB. Because the routine also creates temporary raw MATLAB files, replicators should have at least 20 GB of free storage.

Data requirements

The authors of the manuscript have legitimate access to and permission to use the data used in this manuscript. Some data **cannot be provided** in the public replication package. Data access requires permission to use confidential data from the French Statistical Confidentiality Committee and a subscription to CASD. For information on obtaining permission to access confidential microdata, see:

- https://www.comite-du-secret.fr/
- https://www.casd.eu/en/

Please see the section below for details on which data cannot be made publicly available and how to obtain access to them.

Details on each data source

(1) EAP data (not provided)

We obtain product level data from the Annual Production Survey (Enquete Annuelle de Production, EAP). The EAP is a confidential micro dataset available through CASD. A replicator should request access to EAP for all years from 2009 through 2019. This provides annual SAS microdata files. In the CASD environment, the files have the following structure:

```
Data/EAP_EAP_2009/eap2009.sas (INSEE, 2009a)
Data/EAP_EAP_2010/eap2010.sas (INSEE, 2010a)
Data/EAP_EAP_2011/eap2011.sas (INSEE, 2011a)
Data/EAP_EAP_2012/eap2012.sas (INSEE, 2012a)
Data/EAP_EAP_2013/eap2013.sas (INSEE, 2013a)
Data/EAP_EAP_2014/eap2014.sas (INSEE, 2014a)
Data/EAP_EAP_2015/eap2015.sas (INSEE, 2015a)
Data/EAP_EAP_2016/eap2016.sas (INSEE, 2016a)
Data/EAP_EAP_2017/eap2017.sas (INSEE, 2017a)
Data/EAP_EAP_2018/eap2018.sas (INSEE, 2018a)
Data/EAP_EAP_2019/eap2019.sas (INSEE, 2019a)
```

The following data files are included in the 'theory' replication package and are manual extractions of empirical figures:

```
theory/plotcoordinates/firmsize_user.csv (extracted by the authors from Figure 1)
theory/plotcoordinates/prodcount.csv (extracted by the authors from Figure 1)
theory/plotcoordinates/figure_rank_new_prod_lagNAll_datapoints.csv (extracted by the authors from
Figure 2a)
theory/plotcoordinates/figure_rank_new_prod_lagN1_datapoints.csv (extracted by the authors from
Figure 2b)
theory/plotcoordinates/figure_rank_new_prod_lagN5_datapoints.csv (extracted by the authors from
Figure 2c)
theory/plotcoordinates/figure_rank_new_prod_lagN10_datapoints.csv (extracted by the authors from
Figure 2d)
```

These files are used only as empirical inputs to the theoretical figures. Data was accessed last in August 2026.

(2) FARE data (not provided)

We obtain firm-level income statement and balance sheet information from the Fichier approche des resultats d'Esane (FARE), which covers the universe of French private companies. FARE is a confidential micro dataset available through CASD under the same access conditions as the EAP. A replicator should request access to FARE for 2008 and all years through 2019. In the CASD environment, the annual files are located in folders with the following structure:

```
Data/Statistique annuelle d'entreprise_FARE_2008/tab.sas (INSEE, 2008)
Data/Statistique annuelle d'entreprise_FARE_2009/tab.sas (INSEE, 2009b)
Data/Statistique annuelle d'entreprise_FARE_2010/fare2010.sas (INSEE, 2010b)
Data/Statistique annuelle d'entreprise_FARE_2011/fare2011.sas (INSEE, 2011b)
Data/Statistique annuelle d'entreprise_FARE_2012/fare2012_meth2012.sas (INSEE, 2012b)
Data/Statistique annuelle d'entreprise_FARE_2013/fare2013meth2013.sas (INSEE, 2013b)
Data/Statistique annuelle d'entreprise_FARE_2014/fare2014meth2014.sas (INSEE, 2014b)
Data/Statistique annuelle d'entreprise_FARE_2015/fare2015meth2015.sas (INSEE, 2015b)
Data/Statistique annuelle d'entreprise_FARE_2016/fare2016meth2016.sas (INSEE, 2016b)
Data/Statistique annuelle d'entreprise_FARE_2017/fare2017meth2017.sas (INSEE, 2017b)
Data/Statistique annuelle d'entreprise_FARE_2018/fare2018meth2018.sas (INSEE, 2018b)
Data/Statistique annuelle d'entreprise_FARE_2019/fare2019meth2019.sas (INSEE, 2019b)
```

Data was accessed last in August 2026.

(3) Contours of profiled enterprises (not provided)

We use the contours of profiled enterprises (Contours des entreprises profilees) to consolidate legal units belonging to the same profiled enterprise. The data are confidential and available through CASD. A replicator should request access for 2009 to 2019. In the CASD environment, the annual files are located in folders with the following structure:

```
Data/Statistique annuelle d'entreprise_Contours des entreprises profilées_2009/contour09.sas (INSEE, 2009c)
Data/Statistique annuelle d'entreprise_Contours des entreprises profilées_2010/contour10.sas (INSEE, 2010c)
Data/Statistique annuelle d'entreprise_Contours des entreprises profilées_2011/contour11.sas (INSEE, 2011c)
Data/Statistique annuelle d'entreprise_Contours des entreprises profilées_2012/contour12.sas (INSEE, 2012c)
Data/Statistique annuelle d'entreprise_Contours des entreprises profilées_2013/contour13.sas (INSEE, 2013c)
Data/Statistique annuelle d'entreprise_Contours des entreprises profilées_2014/contour14.sas (INSEE, 2014c)
Data/Statistique annuelle d'entreprise_Contours des entreprises profilées_2015/contour15.sas (INSEE, 2015c)
Data/Statistique annuelle d'entreprise_Contours des entreprises profilées_2016/contour16.sas (INSEE, 2016c)
Data/Statistique annuelle d'entreprise_Contours des entreprises profilées_2017/contour17.sas (INSEE, 2017c)
Data/Statistique annuelle d'entreprise_Contours des entreprises profilées_2018/contour18.sas (INSEE, 2018c)
Data/Statistique annuelle d'entreprise_Contours des entreprises profilées_2019/contour19.sas (INSEE, 2019c)
```

Data was accessed last in August 2026.

(4) DADS Postes data (not provided)

We obtain the number of R&D employees and the number of establishments from the Declaration Annuelle de Donnees Sociales (DADS Postes), a matched employer-employee dataset at the job-spell level. The data are confidential and available through CASD. A replicator should request access to DADS Postes for 2010 to 2019. In the CASD environment, the annual files are located in folders with the structure `Data/DADS_DADS Postes_YYYY`. The internal organization of the raw files changes over time (departmental files up to 2013, regional and Ile-de-France files for 2014 to 2017, and four national files for 2018 and 2019); the SAS script `empirics/code/sas/exportSAS_DADSpostes.sas` handles each of these structures. The complete list of files used, by period, is the following (210 files in total).

For 2010, one file per department group DEP:

```
Data/DADS_DADS Postes_2010/postDEP.sas (INSEE, 2010d)
```

where DEP takes the 28 values: 11aa, 11bb, 11cc, 11dd, 21, 22, 23, 24, 25, 26, 31, 41, 42, 43, 52, 53, 54, 72, 73, 74, 82a, 82b, 83, 91, 93, 94, 97, 99.

For 2011:

```
Data/DADS_DADS Postes_2011/postDEP.sas (INSEE, 2011d)
```

where DEP takes the 24 values: 11, 21, 22, 23, 24, 25, 26, 31, 41, 42, 43, 52, 53, 54, 72, 73, 74, 82, 83, 91, 93, 94, 97, 99.

For 2012 and 2013 (YYYY = 2012, 2013):

```
Data/DADS_DADS Postes_YYYY/postDEP.sas (INSEE, 2012d, 2013d)
Data/DADS_DADS Postes_YYYY/Ile de France/postDEP2.sas (INSEE, 2012d, 2013d)
```

where DEP takes the 23 values: 21, 22, 23, 24, 25, 26, 31, 41, 42, 43, 52, 53, 54, 72, 73, 74, 82, 83, 91, 93, 94, 97, 99; and DEP2 takes the 8 values: 75, 77, 78, 91, 92, 93, 94, 95.

For 2014 to 2017 (YYYY = 2014, ..., 2017):

```
Data/DADS_DADS Postes_YYYY/Régions/postREG.sas (INSEE, 2014d, 2015d, 2016d, 2017d)
Data/DADS_DADS Postes_YYYY/Départements/postDEP2.sas (INSEE, 2014d, 2015d, 2016d, 2017d)
```

where REG takes the 14 values: 24, 27, 28, 32, 44, 52, 53, 75, 76, 84, 93, 94, 97, 99; and DEP2 takes the 8 values: 75, 77, 78, 91, 92, 93, 94, 95.

For 2018 and 2019 (YYYY = 2018, 2019):

```
Data/DADS_DADS Postes_YYYY/post_1.sas (INSEE, 2018d, 2019d)
Data/DADS_DADS Postes_YYYY/post_2.sas (INSEE, 2018d, 2019d)
Data/DADS_DADS Postes_YYYY/post_3.sas (INSEE, 2018d, 2019d)
Data/DADS_DADS Postes_YYYY/post_4.sas (INSEE, 2018d, 2019d)
```

Data was accessed last in August 2026.

(5) LiFi data (not provided)

We determine firms' ownership status (independent or part of a group) from the survey on financial links between enterprises (Liaisons Financieres entre Societes, LiFi). The data are confidential and available through CASD. A replicator should request access to LiFi for 2010 to 2019. In the CASD environment, the annual files are located in folders with the following structure:

```
Data/LIFI_Enquete LIFI_2010/lar.sas (INSEE, 2010e)
Data/LIFI_Enquete LIFI_2011/lar.sas (INSEE, 2011e)
Data/LIFI_LIFI_2012/lar.sas (INSEE, 2012e)
Data/LIFI_LIFI_2013/lar.sas (INSEE, 2013e)
Data/LIFI_LIFI_2014/liaison_reelle.sas (INSEE, 2014e)
Data/LIFI_LIFI_2015/liaison_reelle.sas (INSEE, 2015e)
Data/LIFI_LIFI_2016/liaison_reelle.sas (INSEE, 2016e)
Data/LIFI_LIFI_2017/liaison_reelle.sas (INSEE, 2017e)
Data/LIFI_LIFI_2018/liaison_reelle.sas (INSEE, 2018e)
Data/LIFI_LIFI_2019/liaison_reelle.sas (INSEE, 2019e)
```

In the CASD environment, the 2010 and 2011 folder names contain accented characters ("Enquete" appears as "Enquête").

Data was accessed last in August 2026.

(6) GECIR data (not provided)

We obtain firms' R&D expenditure, R&D employment, R&D wage bill, and R&D tax credit (Credit d'Impot Recherche, CIR) from the GECIR files, which collect the CIR declarations (form 2069A). The data are confidential and available through CASD. The SAS export script converts the annual files for 2008 to 2014 and the 2020, 2023, and 2025 vintages of the declaration files; the analysis then uses the 2010 to 2013 annual files, the 2020 vintage (for the years 2014 to 2018), and the 2023 vintage (for the year 2019). In the CASD environment, the files used are the following:

```
Data/BGCI_GECIR_GECIR/gecir_2008.sas (DESF, 2025)
Data/BGCI_GECIR_GECIR/gecir_2009.sas (DESF, 2025)
Data/BGCI_GECIR_GECIR/gecir_2010.sas (DESF, 2025)
Data/BGCI_GECIR_GECIR/gecir_2011.sas (DESF, 2025)
Data/BGCI_GECIR_GECIR/gecir_2012.sas (DESF, 2025)
Data/BGCI_GECIR_GECIR/gecir_2013.sas (DESF, 2025)
Data/BGCI_GECIR_GECIR/gecir_2014.sas (DESF, 2025)
Data/BGCI_GECIR_GECIR/dec_2069a_092020.sas (DESF, 2025)
Data/BGCI_GECIR_GECIR/d_2069_liasses_082023_diff.sas (DESF, 2025)
Data/BGCI_GECIR_GECIR/dec_2069_052025_diff.sas (DESF, 2025)
```

Data was accessed last in August 2026.

(7) Nomenclature and correspondence tables (provided)

The following non-confidential inputs are provided in the replication package in `empirics/data/external`.

Annual PRODFRA-10 nomenclature files, provided by INSEE:

```
empirics/data/external/concord_pc10/nom_pc10/pc10_2009.csv (INSEE, 2020a)
empirics/data/external/concord_pc10/nom_pc10/pc10_2010.csv (INSEE, 2020a)
empirics/data/external/concord_pc10/nom_pc10/pc10_2011.csv (INSEE, 2020a)
empirics/data/external/concord_pc10/nom_pc10/pc10_2012.csv (INSEE, 2020a)
empirics/data/external/concord_pc10/nom_pc10/pc10_2013.csv (INSEE, 2020a)
empirics/data/external/concord_pc10/nom_pc10/pc10_2014.csv (INSEE, 2020a)
empirics/data/external/concord_pc10/nom_pc10/pc10_2015.csv (INSEE, 2020a)
empirics/data/external/concord_pc10/nom_pc10/pc10_2016.csv (INSEE, 2020a)
empirics/data/external/concord_pc10/nom_pc10/pc10_2017.csv (INSEE, 2020a)
empirics/data/external/concord_pc10/nom_pc10/pc10_2018.csv (INSEE, 2020a)
empirics/data/external/concord_pc10/nom_pc10/pc10_2019.csv (INSEE, 2020a)
empirics/data/external/concord_pc10/nom_pc10/pc10_2020.csv (INSEE, 2020a)
```

Year-to-year correspondence tables between successive PRODFRA-10 vintages, provided by INSEE:

```
empirics/data/external/concord_pc10/tablescorresp/pc10_2010-pc10_2009.xlsx (INSEE, 2020b)
empirics/data/external/concord_pc10/tablescorresp/pc10_2011-pc10_2010.xlsx (INSEE, 2020b)
empirics/data/external/concord_pc10/tablescorresp/pc10_2012-pc10_2011.xlsx (INSEE, 2020b)
empirics/data/external/concord_pc10/tablescorresp/pc10_2013-pc10_2012.xlsx (INSEE, 2020b)
empirics/data/external/concord_pc10/tablescorresp/pc10_2014-pc10_2013.xlsx (INSEE, 2020b)
empirics/data/external/concord_pc10/tablescorresp/pc10_2015-pc10_2014.xlsx (INSEE, 2020b)
empirics/data/external/concord_pc10/tablescorresp/pc10_2016-pc10_2015.xlsx (INSEE, 2020b)
empirics/data/external/concord_pc10/tablescorresp/pc10_2017-pc10_2016.xlsx (INSEE, 2020b)
empirics/data/external/concord_pc10/tablescorresp/pc10_2018-pc10_2017.xlsx (INSEE, 2020b)
empirics/data/external/concord_pc10/tablescorresp/pc10_2019-pc10_2018.xlsx (INSEE, 2020b)
empirics/data/external/concord_pc10/tablescorresp/pc10_2020-pc10_2019.xlsx (INSEE, 2020b)
empirics/data/external/concord_pc10/tablescorresp/pc10_2010-pc10_2009.csv (INSEE, 2020b)
empirics/data/external/concord_pc10/tablescorresp/pc10_2011-pc10_2010.csv (INSEE, 2020b)
empirics/data/external/concord_pc10/tablescorresp/pc10_2012-pc10_2011.csv (INSEE, 2020b)
empirics/data/external/concord_pc10/tablescorresp/pc10_2013-pc10_2012.csv (INSEE, 2020b)
empirics/data/external/concord_pc10/tablescorresp/pc10_2014-pc10_2013.csv (INSEE, 2020b)
empirics/data/external/concord_pc10/tablescorresp/pc10_2015-pc10_2014.csv (INSEE, 2020b)
empirics/data/external/concord_pc10/tablescorresp/pc10_2016-pc10_2015.csv (INSEE, 2020b)
empirics/data/external/concord_pc10/tablescorresp/pc10_2017-pc10_2016.csv (INSEE, 2020b)
empirics/data/external/concord_pc10/tablescorresp/pc10_2018-pc10_2017.csv (INSEE, 2020b)
empirics/data/external/concord_pc10/tablescorresp/pc10_2019-pc10_2018.csv (INSEE, 2020b)
empirics/data/external/concord_pc10/tablescorresp/pc10_2020-pc10_2019.csv (INSEE, 2020b)
```

Correspondence tables between the CN8 and PRODCOM classifications for 2009 to 2019 and the CN8-BEC 2020 correspondence table, both obtained from the Eurostat RAMON archive (https://circabc.europa.eu/ui/group/c1b49c83-24a7-4ff2-951c-621ac0a89fd8):

```
empirics/data/external/corresp_pc8_bec/cn8_pc8corresp/CN 2009 - PRODCOM 2009.csv (Eurostat, 2020)
empirics/data/external/corresp_pc8_bec/cn8_pc8corresp/CN 2010 - PRODCOM 2010.csv (Eurostat, 2020)
empirics/data/external/corresp_pc8_bec/cn8_pc8corresp/CN 2011 - PRODCOM 2011.csv (Eurostat, 2020)
empirics/data/external/corresp_pc8_bec/cn8_pc8corresp/CN 2012 - PRODCOM 2012.csv (Eurostat, 2020)
empirics/data/external/corresp_pc8_bec/cn8_pc8corresp/CN 2013 - PRODCOM 2013.csv (Eurostat, 2020)
empirics/data/external/corresp_pc8_bec/cn8_pc8corresp/CN 2014 - PRODCOM 2014.csv (Eurostat, 2020)
empirics/data/external/corresp_pc8_bec/cn8_pc8corresp/CN 2015 - PRODCOM 2015.csv (Eurostat, 2020)
empirics/data/external/corresp_pc8_bec/cn8_pc8corresp/CN 2016 - PRODCOM 2016.csv (Eurostat, 2020)
empirics/data/external/corresp_pc8_bec/cn8_pc8corresp/CN 2017 - PRODCOM 2017.csv (Eurostat, 2020)
empirics/data/external/corresp_pc8_bec/cn8_pc8corresp/CN 2018 - PRODCOM 2018.csv (Eurostat, 2020)
empirics/data/external/corresp_pc8_bec/cn8_pc8corresp/CN 2019 - PRODCOM 2019.csv (Eurostat, 2020)
empirics/data/external/corresp_pc8_bec/cn8_bec_2020/CN_BEC_2020.xlsx (Eurostat, 2020)
empirics/data/external/corresp_pc8_bec/cn8_bec_2020/CN_BEC_2020.csv (Eurostat, 2020)
```

Product-code correspondence tables accompanying the EAP (ProdEap), provided by INSEE, and the list of firm identifiers belonging to the exhaustive part of the EAP sample (EAP1), provided by INSEE and used to restrict the sample as described in Appendix B.1.1 of the paper:

```
empirics/data/external/INSEE/correspondance tables INSEE/ProdEap_correspondance_table_2009_2017.xls
(INSEE, 2020c)
empirics/data/external/INSEE/correspondance tables INSEE/ProdEap_correspondance_table_2018_2020.xls
(INSEE, 2020c)
empirics/data/external/INSEE/correspondance tables INSEE/ProdEap_correspondance_table_2009_2017.csv
(INSEE, 2020c)
empirics/data/external/INSEE/correspondance tables INSEE/ProdEap_correspondance_table_2018_2020.csv
(INSEE, 2020c)
empirics/data/external/INSEE/EAP_listsirens_bysample.csv (INSEE, 2020d)
```

Dataset list

Provided (non-confidential)

File	Source
`theory/empirics/data/firmsize_user.csv`	Authors (aggregates derived from EAP)
`theory/empirics/data/prodcount.csv`	Authors (aggregates derived from EAP)
`theory/empirics/data/figure_rank_new_prod_lagNAll_datapoints.csv`	Authors (aggregates derived from EAP)
`theory/empirics/data/figure_rank_new_prod_lagN1_datapoints.csv`	Authors (aggregates derived from EAP)
`theory/empirics/data/figure_rank_new_prod_lagN5_datapoints.csv`	Authors (aggregates derived from EAP)
`theory/empirics/data/figure_rank_new_prod_lagN10_datapoints.csv`	Authors (aggregates derived from EAP)
`empirics/data/external/concord_pc10/nom_pc10/pc10_YYYY.csv` (YYYY = 2009-2020, 12 files)	INSEE (Authors' elaboration)
`empirics/data/external/concord_pc10/tablescorresp/pc10_YYYY-pc10_YYYY-1.xlsx` (YYYY = 2010-2020, 11 files)	INSEE (Authors' elaboration)
`empirics/data/external/concord_pc10/tablescorresp/pc10_YYYY-pc10_YYYY-1.csv` (YYYY = 2010-2020, 11 files)	INSEE (Authors' elaboration)
`empirics/data/external/corresp_pc8_bec/cn8_pc8corresp/CN YYYY - PRODCOM YYYY.csv` (YYYY = 2009-2019, 11 files)	Eurostat (RAMON)

File	Source
empirics/data/external/corresp_pc8_bec/cn8_bec_2020/CN_BEC_2020.xlsx	Eurostat (RAMON)
empirics/data/external/corresp_pc8_bec/cn8_bec_2020/CN_BEC_2020.csv	Eurostat (RAMON)
empirics/data/external/INSEE/correspondance tables INSEE/ProdEap_correspondance_table_2009_2017.xls	INSEE
empirics/data/external/INSEE/correspondance tables INSEE/ProdEap_correspondance_table_2018_2020.xls	INSEE
empirics/data/external/INSEE/correspondance tables INSEE/ProdEap_correspondance_table_2009_2017.csv	INSEE
empirics/data/external/INSEE/correspondance tables INSEE/ProdEap_correspondance_table_2018_2020.csv	INSEE
empirics/data/external/INSEE/EAP_listsirens_bysample.csv	INSEE

Not provided (confidential)

These files can only be accessed within the CASD environment. The complete file lists for each source are given in the details on each data source above.

File	Source
Data/EAP_EAP_YYYY/eapYYYY.sas (YYYY = 2009-2019)	INSEE
Data/Statistique annuelle d'entreprise_FARE_YYYY/... (YYYY = 2008-2019)	INSEE
Data/Statistique annuelle d'entreprise_Contours des entreprises profilees_YYYY/contourYY.sas (YYYY = 2009-2019)	INSEE
Data/DADS_DADS Postes_YYYY/... (YYYY = 2010-2019, 210 files)	INSEE
Data/LIFI_Enquete LIFI_YYYY/lar.sas (YYYY = 2010-2011)	INSEE
Data/LIFI_LIFI_YYYY/lar.sas (YYYY = 2012-2013)	INSEE
Data/LIFI_LIFI_YYYY/liaison_reelle.sas (YYYY = 2014-2019)	INSEE
Data/BGCI_GECIR_GECIR/gecir_YYYY.sas (YYYY = 2008-2014)	DESF
Data/BGCI_GECIR_GECIR/dec_2069a_092020.sas	DESF
Data/BGCI_GECIR_GECIR/d_2069_liasses_082023_diff.sas	DESF
Data/BGCI_GECIR_GECIR/dec_2069_052025_diff.sas	DESF

Description of programs/code

Description of theoretical programs

Program `theory/file1_master.m` runs the complete theoretical replication.

Parameters and quantification

- `file2_parameters_targets.m` defines the simulation parameters, externally chosen parameters, calibration targets, and parameter grids.
- `file3_quantification.m` runs the model quantification and selects the preferred calibration.
- `file3a_stepsizesolver.m` solves for the process-innovation step size and its decline with product tenure.
- `file3b_typesolver.m` solves for the innovation-efficiency type parameters and type shares.
- `file3c_xrelsolver.m` solves for relative innovation rates.
- `file3d_innerloop_danial.m` solves the value-function inner loop.
- `file3e_entranttypesolver.m` solves for entrant type shares.
- `file3f_firmsizedistribution.m` computes the stationary firm-size distribution.
- `file3g_growthcalculator.m` calculates aggregate growth.
- `file3h_firmsim_lite.m` runs the simulation used to estimate the product-creation tail moment during quantification.

Simulation and dataset construction

- `file4_simulation.m` simulates the quantified innovation-bursts model and the Klette-Kortum benchmark.
- `file4a_firmsim_ces.m` is the firm-level simulation routine called by `file4_simulation.m`.
- `file5_calibrate_noise.m` calibrates the standard deviation of idiosyncratic product-quality shocks.
- `file5a_decomposition_shapley.m` performs the Shapley-Owen decomposition used by the noise calibration.
- `file6_create_datasets.m` converts the raw MATLAB simulation output into `simulation_firmlevel_XX.csv` and `simulation_productlevel_XX.csv`, where `XX` identifies the Monte Carlo draw.
- `hurwitzZeta_danial.m` and `randraw.m` are numerical helper routines.

Theoretical tables and figures

- `file7_output_parameter_table.m` produces Table V.
- `file7a_policy_findbaselineRDcost.m` calculates the R&D cost scalars reported in Table V.
- `file8_output_targeted_moments.m` produces Table VI.
- `file9_plot_firmsizedistributions.m` and `file9a_plot_firmsizedistributions.m` produce the firm-size CSV inputs for Figures 9 and 10.
- `stata_file1_master.do` calls all Stata figure and table routines.
- `stata_file2_figure9.do` produces Figure 9.
- `stata_file3_figure10.do` produces Figure 10.
- `stata_file4_figure11.do` produces Figure 11 and Appendix Figure C13.
- `stata_file5_figure12.do` produces Figure 12.
- `stata_file6_intext_agenumbers.do` produces the simulated firm-age moments reported in the text.
- `stata_file7_table7_model.do` produces the model results for Table VII.
- `stata_file8_table8_model.do` produces the model results for Table VIII.

- `file10_plot_typeshare.m` produces Figure 13.
- `file11_plot_growtcontribution.m` produces Figure 14.

Description of empirical programs

Program `empirics/code/master.do` sets up the environment (packages, seed, graph scheme, paths, sample period, and the confidentiality flag `conf_check`), creates the folder structure, and runs all programs below in the order in which they are listed.

SAS export programs

The six scripts in `empirics/code/sas` convert the raw SAS microdata in the CASD data folder into Stata datasets saved in `empirics/data/raw`. They are called by the master file through Windows `shell` commands, with the raw-data and destination folders passed as arguments.

- `exportSAS_eap.sas` converts the annual EAP files for 2009 to 2019.
- `exportSAS_fare.sas` converts the annual FARE files for 2008 to 2019, keeping only the required variables.
- `exportSAS_contours.sas` converts the annual contours of profiled enterprises for 2009 to 2019.
- `exportSAS_DADSpostes.sas` converts the DADS Postes files for 2010 to 2019, keeping only the required variables and handling the changing structure of the raw files across years.
- `exportSAS_LiFi.sas` converts the annual LiFi files for 2010 to 2019.
- `exportSAS_gecir.sas` converts the GECIR files (2008 to 2014 and the 2020, 2023, and 2025 vintages).

Dataset-construction programs

The programs in `empirics/code/data` build the analysis datasets, which are saved in `empirics/data/processed`.

- `contours/prepContour_master.do` builds a panel of the contours of profiled enterprises, used to consolidate legal units into firms.
- `corresp_pc8_bec/prepCorres_pc8_bec_master.do` builds a correspondence between PRODCOM-8 codes and the BEC classification (via CN8), used to identify consumer goods; it calls `cn8_pc8_input_files.do`.
- `concord_pc10/prepConcord_pc10_master.do` constructs time-consistent PRODFRA-10 product codes using the connected-components concordance described in Appendix B.1.3 of the paper; it calls `corresp_stata.do` and `input_file_PC_over_time_edited.do`. The code is adapted from Bergounhon, Lenoir, and Mejean (2018).
- `fare/prepFARE_master.do` cleans and appends the annual FARE files, generates the accounting variables, and consolidates profiled enterprises; it calls `cleanallyears.do`, `genvars.do`, and `consolidation_fare.do`.
- `lifi/prepLiFi_master.do` builds a panel of group structures from the LiFi files, used to determine firms' ownership status.
- `gecir/prepGECIR_rd.do` builds a firm-level panel of R&D expenditure, R&D employment, R&D wage bill, and R&D tax credit from the GECIR files.
- `dads/prepDADS.do` builds firm-level counts of R&D employees and establishments from the DADS Postes files for the EAP firms; it calls `consolidation_dads.do`.
- `eap/1_EAPpanel_preprawdata_firmprod.do` imports and cleans the annual EAP files, appends them into a firm-product panel, consolidates historical groups, and concords products over time. The panel is constructed at five levels of product aggregation (concorded PRODFRA-10, raw PRODFRA-10, PRODCOM-8, PRODCOM-6, and PRODCOM-4).
- `eap/2_EAPpanel_prepsample_firmprod.do` restricts the panel to manufacturing firms matched to FARE that belong to the exhaustive part of the EAP sample.

- `eap/3_1_EAPpanel_prepworkdata_firm.do` builds the firm-level analysis dataset: it deflates values, cleans firms' re-entry into the survey, and generates entry and exit variables.
- `eap/3_1_1_EAPpanel_prepworkdata_mode3_firm.do` builds the firm-level dataset using only production mode 3, used in Figure C8.
- `eap/3_1_2_EAPpanel_prepworkdata_noreturnprod_firm.do` builds the firm-level dataset excluding product re-entry, used in Figure C10.
- `eap/3_2_EAPpanel_prepworkdata_firmprod.do` builds the firm-product-level analysis dataset.

Empirical tables and figures

Each program in `empirics/code/analysis` is named after the exhibit it produces and reads the analysis datasets in `empirics/data/processed`. The mapping from exhibits to programs and output files is given in the list of tables and figures below.

- `Table_1.do` produces Table I (summary statistics).
- `Figure_1a.do` and `Figure_1b.do` produce the two panels of Figure 1 (decomposing firm concentration into products and revenue per product).
- `Table_2.do` produces the top and bottom panels of Table II (product creation and destruction).
- `Figure_2.do` produces the four panels of Figure 2 (distribution of the number of new products, for all firms and by initial product count).
- `Figure_3.do` produces Figure 3 (average revenue per new product).
- `Figure_4.do` produces Figure 4 (distribution of the number of new products by level of aggregation).
- `Figure_5.do` produces Figure 5 (distribution of the number of new products by proxies of fixed firm boundary).
- `Table_3.do` produces Table III (contribution of product innovation and destruction to revenue growth).
- `Figure_6.do` produces Figure 6 (creative destruction and aggregate revenue growth).
- `Figure_7.do` produces Figure 7 (firm-level persistence in average revenue from new products).
- `Figure_8.do` produces Figure 8 (life cycle of revenue growth).
- `Table_7.do` produces the data columns of Table VII.
- `Table_8.do` produces the data columns of Table VIII.
- `Table_intextnumbers.do` produces the empirical in-text numbers reported in the paper.
- `Table_c1.do` to `Table_c6.do` produce Appendix Tables C1 to C6.
- `Figure_c1.do` to `Figure_c12.do` produce Appendix Figures C1 to C12.
- `Table_d1.do`, `Figure_d1.do`, and `Figure_d2.do` produce Appendix Table D1 and Appendix Figures D1 and D2.
- `_rank_byvar.ado` is a helper program, provided with the package, that computes rank-share distributions and implements the confidentiality binning used in the distribution plots.
- `rename_output_files.do` renames the panels of the multi-panel figures to the final file names listed below.

List of tables and figures

The provided code reproduces all tables and figures in the paper. The table below identifies the program in which each table or figure is created, following the sequence in which the exhibits appear in the paper. Programs with paths starting with `empirics/` require the confidential microdata and can only be run in the CASD environment. Table IV summarizes the model equations and Table B1 presents examples of product codes; neither is produced by code.

Figure/Table #	Program	Output file	Note
Table I	empirics/code/analysis/Table_1.do	empirics/output/table_1.tex	Summary statistics
Figure 1a	empirics/code/analysis/Figure_1a.do	empirics/output/figure_1a.pdf	Firm-concentration decomposition, panel (a)
Figure 1b	empirics/code/analysis/Figure_1b.do	empirics/output/figure_1b.pdf	Firm-concentration decomposition, panel (b)
Table II	empirics/code/analysis/Table_2.do	empirics/output/table_2_top.tex; empirics/output/table_2_bottom.tex	Product creation and destruction
Figure 2	empirics/code/analysis/Figure_2.do	empirics/output/figure_2a.pdf to figure_2d.pdf	Distribution of number of new products
Figure 3	empirics/code/analysis/Figure_3.do	empirics/output/figure_3a.pdf; empirics/output/figure_3b.pdf	Average revenue per new product
Figure 4	empirics/code/analysis/Figure_4.do	empirics/output/figure_4a.pdf; empirics/output/figure_4b.pdf	Distribution of new products by level of aggregation
Figure 5	empirics/code/analysis/Figure_5.do	empirics/output/figure_5a.pdf; empirics/output/figure_5b.pdf	Distribution of new products by proxies of fixed firm boundary
Table III	empirics/code/analysis/Table_3.do	empirics/output/table_3.tex	Contribution of product innovation and destruction to revenue growth
Figure 6	empirics/code/analysis/Figure_6.do	empirics/output/figure_6.pdf	Creative destruction and aggregate revenue growth
Figure 7	empirics/code/analysis/Figure_7.do	empirics/output/figure_7a.pdf; empirics/output/figure_7b.pdf	Persistence in average revenue from new products
Figure 8	empirics/code/analysis/Figure_8.do	empirics/output/figure_8.pdf	Life cycle of revenue growth
Table V	theory/file7_output_parameter_table.m	theory/plots/Table_5.tex	Summary of parameter values
Table VI	theory/file8_output_targeted_moments.m	theory/plots/Table_6.tex	Targeted moments and related parameters
Figure 9a	theory/file9*.m, theory/stata_file2_figure9.do	theory/plots/Figure_9a.pdf	Full firm-size distribution
Figure 9b	theory/file9*.m, theory/stata_file2_figure9.do	theory/plots/Figure_9b.pdf	Firms with low average innovation step size
Figure 9c	theory/file9*.m, theory/stata_file2_figure9.do	theory/plots/Figure_9c.pdf	Firms with high average innovation step size
Figure 10	theory/file9*.m, theory/stata_file3_figure10.do	theory/plots/Figure_10.pdf	Firm-size distribution with Klette-Kortum firm types
Figure 11a	theory/stata_file4_figure11.do	theory/plots/Figure_11_a.pdf	Distribution of new products, all firms
Figure 11b	theory/stata_file4_figure11.do	theory/plots/Figure_11_b.pdf	Distribution of new products, initial product count 1
Table VII	empirics/code/analysis/Table_7.do	empirics/output/table_7.tex	Produces the data columns
Table VII	theory/stata_file7_table7_model.do	theory/plots/table_7_model.csv	Produces the model columns
Table VIII	empirics/code/analysis/Table_8.do	empirics/output/table_8.xlsx	Produces the data columns
Table VIII	theory/stata_file8_table8_model.do	theory/plots/table_8_model.csv	Produces the model columns
Figure 12	theory/stata_file5_figure12.do	theory/plots/figure_12.pdf	Life cycle of revenue growth

Figure/Table #	Program	Output file	Note
Figure 13	theory/file10_plot_typeshare.m	theory/plots/Figure_13.pdf	Size and innovation-efficiency relationship
Figure 14	theory/file11_plot_growtcontribution.m	theory/plots/Figure_14.pdf	Contribution of large firms to productivity growth
In-text empirical numbers	empirics/code/analysis/Table_intextnumbers.do	empirics/output/table_intextnumbers.xlsx	Empirical moments reported in the text
In-text firm-age moments	theory/stata_file6_intext_agenumbers.do	theory/plots/table_age_largefirms_mcavg.tex; theory/plots/table_age_largefirms_mcavg.csv	Simulated average ages reported in Section 5.3
Table C1	empirics/code/analysis/Table_c1.do	empirics/output/table_c1.tex	Product creation rates and firm size
Table C2	empirics/code/analysis/Table_c2.do	empirics/output/table_c2.tex	Alternative decompositions of revenue growth
Table C3	empirics/code/analysis/Table_c3.do	empirics/output/table_c3.tex	Tail parameters for key variables
Table C4	empirics/code/analysis/Table_c4.do	empirics/output/Table_c4.tex	Persistence in revenue per new product
Figure C1	empirics/code/analysis/Figure_c1.do	empirics/output/figure_c1a.pdf; empirics/output/figure_c1b.pdf	Product creation rate by number of products, with and without industry effects
Figure C2	empirics/code/analysis/Figure_c2.do	empirics/output/figure_c2a.pdf; empirics/output/figure_c2b.pdf	Average revenue per new product, different sizes
Figure C3	empirics/code/analysis/Figure_c3.do	empirics/output/figure_c3a.pdf; empirics/output/figure_c3b.pdf	Distribution by level of aggregation, different sizes
Figure C4	empirics/code/analysis/Figure_c4.do	empirics/output/figure_c4a.pdf; empirics/output/figure_c4b.pdf	Distribution by proxies of fixed firm boundary, all firms
Figure C5	empirics/code/analysis/Figure_c5.do	empirics/output/figure_c5a.pdf; empirics/output/figure_c5b.pdf	Distribution with fixed intangible capital
Figure C6	empirics/code/analysis/Figure_c6.do	empirics/output/figure_c6a.pdf; empirics/output/figure_c6b.pdf	Distribution by age
Figure C7	empirics/code/analysis/Figure_c7.do	empirics/output/figure_c7a.pdf; empirics/output/figure_c7b.pdf	Distribution by innovation history
Figure C8	empirics/code/analysis/Figure_c8.do	empirics/output/figure_c8a.pdf; empirics/output/figure_c8b.pdf	Distribution by production mode
Figure C9	empirics/code/analysis/Figure_c9.do	empirics/output/figure_c9a.pdf; empirics/output/figure_c9b.pdf	Distribution, consumer goods
Figure C10	empirics/code/analysis/Figure_c10.do	empirics/output/figure_c10a.pdf; empirics/output/figure_c10b.pdf	Distribution excluding product re-entry
Figure C11	empirics/code/analysis/Figure_c11.do	empirics/output/figure_c11a.pdf to figure_c11f.pdf	Probability that production ends by size of innovation burst
Table C5	empirics/code/analysis/Table_c5.do	empirics/output/table_c5.tex	Cannibalization: probability of product exit
Table C6	empirics/code/analysis/Table_c6.do	empirics/output/table_c6.tex	Cannibalization: continuing products
Figure C12	empirics/code/analysis/Figure_c12.do	empirics/output/figure_c12.pdf	Life cycle of price growth
Figure C13a	theory/stata_file4_figure11.do	theory/plots/Figure_c13_a.pdf	Distribution of new products, initial product count 5

Figure/Table #	Program	Output file	Note
Figure C13b	`theory/stata_file4_figure11.do`	`theory/plots/Figure_c13_b.pdf`	Distribution of new products, initial product count 10
Table D1	`empirics/code/analysis/Table_d1.do`	`empirics/output/table_d1_top.tex`; `empirics/output/table_d1_bottom.tex`	R&D spending and firm product count
Figure D1	`empirics/code/analysis/Figure_d1.do`	`empirics/output/figure_d1a.pdf`; `empirics/output/figure_d1b.pdf`	R&D spending per product and firm size
Figure D2	`empirics/code/analysis/Figure_d2.do`	`empirics/output/figure_d2a.pdf`; `empirics/output/figure_d2b.pdf`	R&D spending per product and number of new products

The following are intermediate MATLAB-to-Stata inputs rather than final paper exhibits:

```
theory/plots/theory_firmsize_stata.csv
theory/plots/theory_firmsize_homogeneous_stata.csv
```

References

Data references

Where CASD has released more than one version of a vintage, the DOI of the most recent version is cited. For DADS Postes, each year's DOI covers all the departmental, regional, or national files of that year; the single GECIR DOI covers all the GECIR files.

- Insee (2009a), Enquete Annuelle de Production. Accessed through CASD project MULTIEX between 2019 and 2026. https://doi.org/10.34724/CASD.40.121.V1
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